HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

Student No.: 07 Name: _____ Marks: / 46

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let a be a constant. If the quadratic equation $x^2 + ax + a = 1$ has equal roots, then $a =$

   A. -1.

   B. 2.

   C. 0 or -4.

   D. 0 or 4.

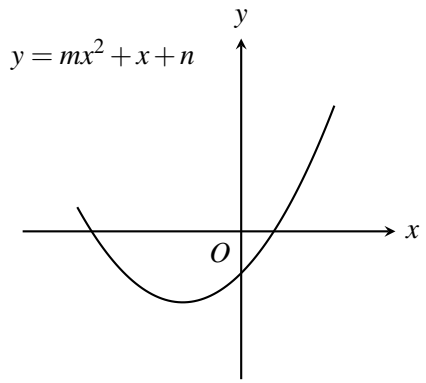
(DSE-2014 PII Q04)

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2. The figure shows the graph of $y = mx^2 + x + n$, where m and n are constants. Which of the following is true?



A. $m < 0$ and $n < 0$

B. $m < 0$ and $n > 0$

C. $m > 0$ and $n < 0$

D. $m > 0$ and $n > 0$

(DSE-2014 PII Q05)

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3. If $a > b$ and $k < 0$, which of the following must be true?

I. $a^2 > b^2$

II. $a + k > b + k$

III. $\frac{a}{k^2} > \frac{b}{k^2}$

A. I only

B. II only

C. I and III only

D. II and III only

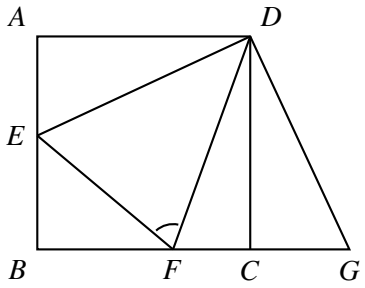
(DSE-2014 PII Q06)

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4. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

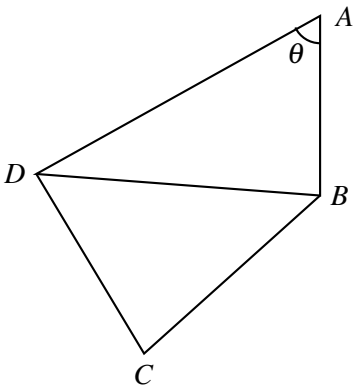
(DSE-2014 PII Q16)

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5. In the figure, $\angle ABD = \angle ADC = \angle BCD = 90°$. If $AB = \ell$, then $CD =$



A. $\ell \sin\theta$.

B. $\ell \cos\theta$.

C. $\ell \sin\theta \tan\theta$.

D. $\dfrac{\ell \tan\theta}{\cos\theta}$.

(DSE-2014 PII Q18)

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6. $\big(\cos(90^\circ+\theta)+1\big)\big(\sin(360^\circ-\theta)-1\big) =$

A. $-\cos^2\theta$.

B. $-\sin^2\theta$.

C. $\cos^2\theta$.

D. $\sin^2\theta$.

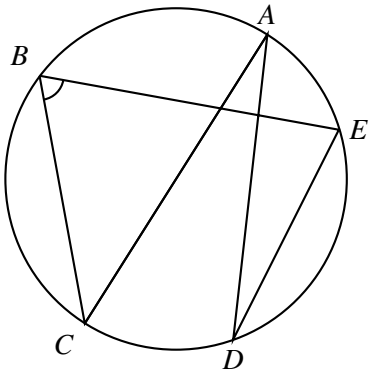
(DSE-2014 PII Q19)

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7. In the figure, AC is a diameter of the circle $ABCDE$. If $\angle ADE = 28°$, then $\angle CBE =$



A. 56°.

B. 62°.

C. 72°.

D. 76°.

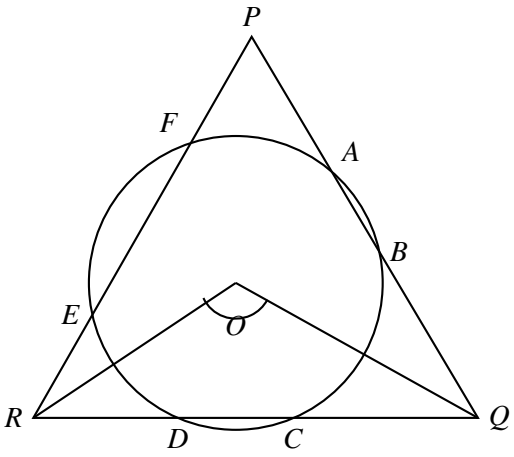
(DSE-2014 PII Q20)

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8. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

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9. If an interior angle of a regular n-sided polygon is greater than an exterior angle by 100°, which of the following are true?

I. The value of n is 10.

II. Each exterior angle of the polygon is 40°.

III. The number of axes of reflectional symmetry of the polygon is 9.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q22)

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10. The rectangular coordinates of the point P are $(-1, \sqrt{3})$. If P is reflected with respect to the x-axis, then the polar coordinates of its image are

A. $(2, 210°)$.

B. $(2, 240°)$.

C. $(4, 210°)$.

D. $(4, 240°)$.

(DSE-2014 PII Q23)

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11. If a diameter of the circle $x^2+y^2-8x+ky-214=0$ passes through the point $(6,-5)$ and the slope of the diameter is -4, then $k=$

A. -6.

B. -4.

C. 13.

D. 70.

(DSE-2014 PII Q26)

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12. A box contains m yellow balls and 20 black balls. If a ball is randomly drawn from the box, then the probability of drawing a yellow ball is $\frac{1}{m}$. Find the value of m.

A. 4

B. 5

C. 15

D. 25

(DSE-2014 PII Q27)

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13. The H.C.F. of $3x^4y^2z$, $4xy^5z$ and $6x^2y^3$ is

A. xy^2.

B. xy^2z.

C. $12x^4y^5z$.

D. $12x^7y^9z^2$.

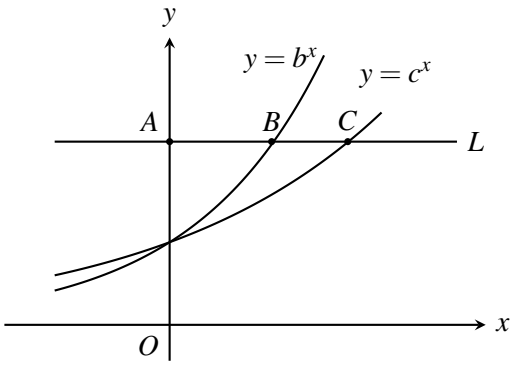
(DSE-2014 PII Q31)

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14. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where b and c are positive constants. If a horizontal line L cuts the y-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at A, B and C respectively, which of the following are true?



I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q32)

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15. Which of the following is the greatest?

A. 124^{241}

B. 241^{214}

C. 412^{142}

D. 421^{124}

(DSE-2014 PII Q33)

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16. $7 \times 2^{10} + 2^8 + 5 \times 2^3 - 2^3 =$

A. 111010100000_2.

B. 111100010000_2.

C. 1110100100000_2.

D. 1111000010000_2.

(DSE-2014 PII Q34)

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17. If β is a real number, then $\dfrac{\beta^2+4}{\beta+2i} =$

A. $\beta-2i$.

B. $\beta+2i$.

C. $2-\beta i$.

D. $2+\beta i$.

(DSE-2014 PII Q36)

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18. If $m > 1$, which of the following are geometric sequences?

I. $2^m, 2^{2m}, 2^{3m}, 2^{4m}$

II. $m, 2m^2, 3m^4, 4m^8$

III. $\log m, \log m^2, \log m^4, \log m^8$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

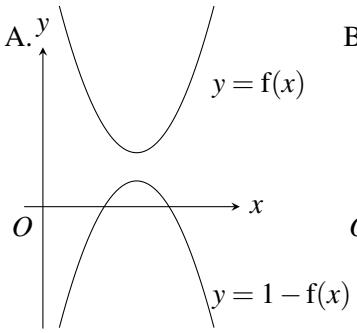
(DSE-2014 PII Q37)

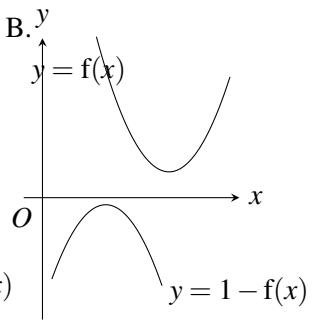
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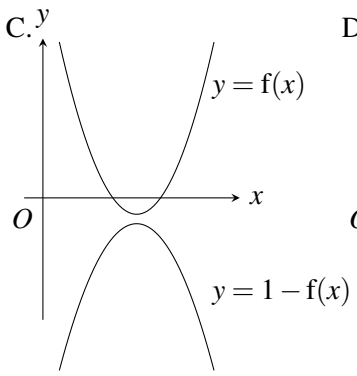
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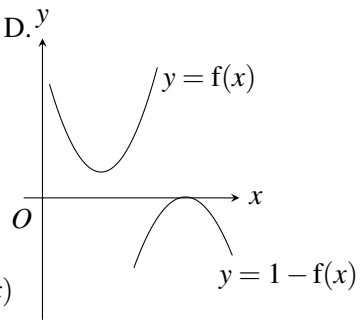
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19. Which of the following may represent the graph of $y = f(x)$ and the graph of $y = 1 - f(x)$ on the same rectangular coordinate system?

A.


B.


C.


D.


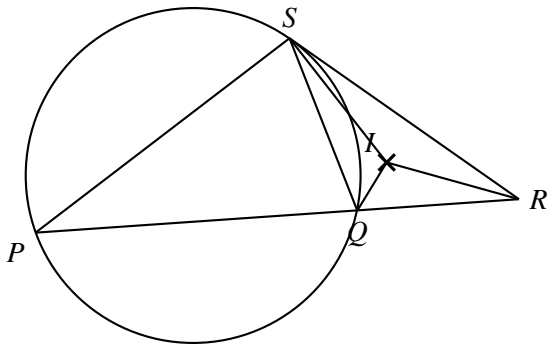
(DSE-2014 PII Q38)

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20. In the figure, PQS is a circle. PQ is produced to R such that RS is the tangent to the circle at S. I is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$



A. 24°.

B. 37°.

C. 43°.

D. 62°.

(DSE-2014 PII Q41)

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21. If the straight line $x - y = k$ and the circle $x^2 + y^2 + 2x - 4y - 1 = 0$ intersect at A and B, then the x-coordinate of the mid-point of AB is

A. $1 + k$.

B. $1 - k$.

C. $\frac{1+k}{2}$.

D. $\frac{1-k}{2}$.

(DSE-2014 PII Q42)

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22. There are 13 boys and 17 girls in a class. If a team of 2 boys and 3 girls is selected from the class to participate in a voluntary service, how many different teams can be formed?

A. 38896

B. 53040

C. 142506

D. 636480

(DSE-2014 PII Q43)

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23. If the variance of the four numbers a, b, c and d is 9, then the variance of the four numbers $14-a$, $14-b$, $14-c$ and $14-d$ is

A. 5.

B. 9.

C. 23.

D. 121.

(DSE-2014 PII Q45)

I can finish _____ questions

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